DICTO Spheroid-13 RD

A true rhombic dodecahedron wrapped by twelve keyed pieces: a nearly round block of thirteen that fills space by translation alone

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Version 1 · 13 October 2026 · DOI 10.5281/zenodo.23332822 · Licence CC BY 4.0 · Data, scripts, renders, nets and checks: studies/2026-10-13-rd13-block at https://github.com/DICTOR-Master/kaleidohedra.
The question whether the rhombic dodecahedron's thirteen fills space as the DICTO Hexa does, the rounded, keyed quality of the block, the true rhombic dodecahedron at its centre, the choice of thicknesses and the name DICTO Spheroid-13 RD are DICTO's. Computation and checking with Claude (Anthropic).

Abstract. A rhombic dodecahedron with its twelve face neighbours, thirteen cells of the face-centred cubic (FCC) honeycomb, fills space by translation alone: of the 183 sublattices of index 13 in the FCC lattice, exactly 8 have the thirteen lattice points as a complete set of representatives, and the 8 are turns of one arrangement, with block translations of length √10 (neighbour spacing √2). Keeping that tiling, we reshape the block: the centre stays a true rhombic dodecahedron, only larger, and the twelve cells around it are moved, corner class by corner class and the same in every block, with every face flat and every cell convex with a minimum thickness, to make the block's outer surface as small as possible. The result, the DICTO Spheroid-13 RD, is nearly round while its outside stays keyed: at minimum thickness 0.15 its roundness $36\pi V^2/S^3$ is 0.707 (the plain block 0.365, a single rhombic dodecahedron 0.741, and the block lattice's Voronoi cell, the roundest single tile with these translations, 0.748), with a centre 1.525 times the size of the lattice's own cells. The block has the tiling's full symmetry (order 6), so it uses three piece shapes: the centre and two outer kinds of six, each three pieces and their mirror images. Each block meets 14 others across all 84 of its outer faces; the centres never meet. Four thicknesses are given (0.08 to 0.35), with printable nets at one scale per set; as a variant, the centre left free to reshape gains at most 0.01 in roundness. The block tiling itself is known: it is the radius-one perfect code of the A_3 lattice (Kovačević and Tan 2018), written explicitly as $x_1 + 3x_2 + 9x_3 \equiv 0 \pmod{13}$ by Kovačević (2026), and its weights 0, 1, 3, 9 are Singer's perfect difference set of the projective plane of order 3. What is new here is the block as thirteen rhombic dodecahedra, rounded and keyed around a true rhombic dodecahedron, with its pieces and nets.

1. The thirteen fills space

Take the FCC lattice as the integer points with even coordinate sum; neighbours are √2 apart and each Voronoi cell, a rhombic dodecahedron, has volume 2. A block of 13 cells, a centre and its twelve neighbours, fills space by translations exactly when its 13 lattice points are a complete set of representatives of a sublattice of index 13. Since 13 is prime, every such sublattice is the kernel of $x \mapsto w \cdot x \bmod 13$ for some $w \neq 0$ (in lattice coordinates), 183 in all up to scale; for exactly 8 the thirteen values are distinct. The 8 sublattices are carried into one another by the 24 turns of the cube, so the tiling is one arrangement in eight orientations, with no left- and right-handed forms. One of them has basis (−3, 0, 1), (0, −1, 3), (1, −3, 0) and Gram matrix [[10, 3, −3], [3, 10, 3], [−3, 3, 10]], covolume 26; its shortest vectors are 6 of length √10, then 2 of √12 (along a three-fold axis) and 6 of √14. The cube symmetries that keep it, the point group of the tiling, have order 6: three turns about that axis and the

inversion. It is the three-dimensional counterpart of the seven-hexagon flower tiling of the plane.

This tiling is known in coding theory. The FCC lattice is the root lattice A_3, and a lattice tiling by a point and its nearest neighbours is a radius-one perfect code: Kovačević and Tan (2018) construct such tilings of A_n in every prime-power dimension, and Kovačević (2026) gives the three-dimensional one explicitly as $\{x : x_1 + 3x_2 + 9x_3 \equiv 0 \pmod{13}\}$, our block lattice ($w = (1, 3, 9)$ in the coordinates used here). The weights explain why it works: 0, 1, 3, 9 is Singer's perfect difference set modulo 13 (Singer 1938), from the projective plane of order 3 with its 13 points and 13 lines; every nonzero residue is a difference of two of them exactly once, so the thirteen lattice points of a block fall into the thirteen classes one each.

2. A true rhombic dodecahedron at the heart

DICTO specified the block's rounded quality: a block that reads as one rounded form, as the DICTO Hexa reads as one block, with its outside still keyed, and with a true rhombic dodecahedron at its centre. To realise it, every corner of the honeycomb was given a displacement depending only on its class modulo the block lattice (39 classes), so a corner shared by two blocks moves once and the blocks still tile exactly. The centre's fourteen corners are not free: they follow from two numbers, its size s and its middle, so it stays an exact rhombic dodecahedron, s times a lattice cell. Constraints: every face stays flat (four corners coplanar), and every cell stays convex with each corner at least a set thickness inside the plane of each face not containing it. Objective: the block's outer surface, which at fixed volume 26 is the same as maximising roundness. Sequential quadratic programming from five small random starts reached the same block at every thickness.

Minimum thickness	Outer surface	Roundness	Centre	Centre volume	Outer volumes (A / B)	Of its hull
plain	59.3970	0.365	× 1	2	2 / 2	0.672
0.08	47.0285	0.735	× 1.589	8.03	1.72 / 1.28	—
0.15	47.6397	0.707	× 1.525	7.10	1.78 / 1.37	0.886
0.25	48.8894	0.654	× 1.433	5.89	1.85 / 1.51	—
0.35	50.5260	0.593	× 1.342	4.84	1.90 / 1.63	0.793

For comparison, the block lattice's Voronoi cell, the roundest convex tile with these translations, has 14 faces, surface 46.7464 and roundness 0.748. At thickness 0.08 the keyed block of thirteen flat-faced cells around a true rhombic dodecahedron comes within 0.6% of its surface. Thicker outer pieces cost roundness and shrink the centre; 0.15 (the main block) and 0.35 (the fattest outer pieces) are the two DICTO chose.

3. The pieces

Every piece keeps the rhombic dodecahedron's combinatorics, 14 corners and 12 four-sided faces meeting as in the rhombic dodecahedron. The centre is exactly a rhombic dodecahedron, its dihedral angles all 120°; in the twelve outer pieces the faces become general flat quadrilaterals and the dihedral angles spread. The block has the tiling's full

symmetry (order 6), so the twelve outer cells fall into two kinds of six, and in each kind the inversion pairs three pieces with their three mirror images.

Thickness	Piece	Count	Volume	Surface	Roundness	Edges	Dihedral angles
0.15	centre (rhombic dodecahedron)	1	7.0959	19.7387	0.740	1.321	120°
	outer A	3 + 3 mirrored	1.7801	8.5047	0.583	0.411–1.321	88.7°–163.4°
	outer B	3 + 3 mirrored	1.3706	7.5322	0.497	0.497–1.321	78.1°–162.3°
0.35	centre (rhombic dodecahedron)	1	4.8351	15.2845	0.740	1.162	120°
	outer A	3 + 3 mirrored	1.8952	8.4204	0.680	0.570–1.162	100.8°–147.0°
	outer B	3 + 3 mirrored	1.6323	7.7655	0.644	0.637–1.162	95.5°–146.0°

4. The centre, the thirteenth block

Each block of thirteen fills space by translation on its own, and at its heart sits a true rhombic dodecahedron, 1.525 times a lattice cell at thickness 0.15 (3.55 cells' volume), turned as the lattice's cells are. It is wrapped on all twelve faces by the twelve, and never meets another block's centre: each centre is an island in the honeycomb of blocks. The twelve outer pieces do all the rounding. As a true rhombic dodecahedron the centre is itself one of Fedorov's five parallelohedra (Fedorov 1885; Grünbaum 2010): taken out, it would fill space on its own too, as the FCC honeycomb at 1.525 times the scale. So the Spheroid-13 RD holds two space-fillings, one inside the other: the block of thirteen, and at its heart the cell the whole lattice is made of.

The free centre (variant). If the centre too is left free to reshape, the block gains at most 0.01 in roundness (0.739, 0.715, 0.664, 0.600 at the four thicknesses, against 0.735, 0.707, 0.654, 0.593), and the centre stays close to a rhombic dodecahedron without being one: at thickness 0.15 it holds 7.18 (3.59 cells' volume), has two kinds of face, six each, and eight of its fourteen corners lie within 1% of one sphere, the other six 18% further in where the outer pieces press. It is centrally symmetric, but its faces miss being parallelograms by about 1.5% of an edge, so taken out of its block it is not one of Fedorov's parallelohedra, unlike the true rhombic dodecahedron of the main form. The four free centres form a graded set (Figure 6). The true rhombic dodecahedron is the main form: simpler to state, to make and to recognise, for almost no loss.

5. How the blocks meet

Inside a block 36 faces are shared: the centre's 12 with the twelve, and 24 among the twelve (the cuboctahedron's edges). All 84 outer faces of a block lie against neighbouring blocks, none left open: 54 against the six nearest blocks (√10 away), 12 against the two along the three-fold axis (√12) and 18 against six more (√14). So each block meets 14 others, the same neighbours as the lattice's 14-faced Voronoi cell. All 84 outer faces belong to outer pieces: the outer pieces of all blocks form one connected keyed network around the islands of the centres.

6. Making it

Printable A4 nets are included for the centre and both outer kinds, with their mirror images: two main sets with the true rhombic dodecahedron at the centre (thickness 0.15 at 28.5 mm per unit, a block about 12.5 cm across; thickness 0.35 at 32.4 mm, about 14.4 cm) and the free-centre variant (0.15 at 24.0 mm, about 10.4 cm), fifteen sheets, each set at one scale so its pieces fit together. Cut lines are solid, fold lines dashed, glued edges numbered alike, with tabs. In the Polyhedraverse app (DICTO's pieces) each version is a kit built from the centre out with outlines showing the next piece; the Kaleidohedra app shows the block whole, pulled apart, and as seven blocks filling space.

7. Checks

The 13 lattice points are distinct modulo each of the 8 sublattices and coincide modulo the other 175. In every block, faces are flat to 10^{-13} and the thirteen volumes sum to 26 exactly; the centre's corners are exactly s times a lattice cell's about its middle; every corner class moves once, so blocks tile; the block is invariant under the six symmetries; all 84 outer faces are matched by faces of neighbouring blocks. Five starts agree at each thickness; 40 further starts with random nudges three to seven times larger, at thickness 0.15 and 0.35, gave the same block every one of the 27 times they converged (the other 13, started far outside the constraints, did not converge), so no rounder valid block was found. Every kit placement reproduces its piece to 10^{-4}, and every kit builds to completion following its outlines. Scripts: round.py (with the rdcentre mode), analysis.py, pieces.py, centre.py, compare.py and nets.mjs in studies/2026-10-13-rd13-block.

8. Relation to known work

A tiling of a lattice by translates of a point and its nearest neighbours is a radius-one perfect code. In the cube lattice such codes exist in every dimension (Golomb and Welch 1970; product constructions in Etzion 2011; Campello et al. 2015 on the lp metrics), while radius two fails in three dimensions (Gravier, Mollard and Payan 1998). In the A_3 = FCC lattice the radius-one code, the 13-cell block tiling of section 1, is due to Kovačević and Tan (2018), given explicitly by Kovačević (2026), and rests on Singer's difference set (1938); see also Dejter, Fuentes and Araujo (2025) on perfect codes in lattice graphs. The count of 8 sublattices and their equivalence under the cube's turns are checked here. Twelve equal spheres round one is the kissing configuration of the FCC packing (Schütte and van der Waerden 1952 for the bound). Parallelohedra, the convex solids that tile by translation, were classified by Fedorov (1885); see Grünbaum (2010). We found no description of rounding a space-filling cluster with flat faces while keeping it keyed and its centre a true rhombic dodecahedron, of the DICTO Spheroid-13 RD, or of its pieces. The search is not exhaustive.

Figures

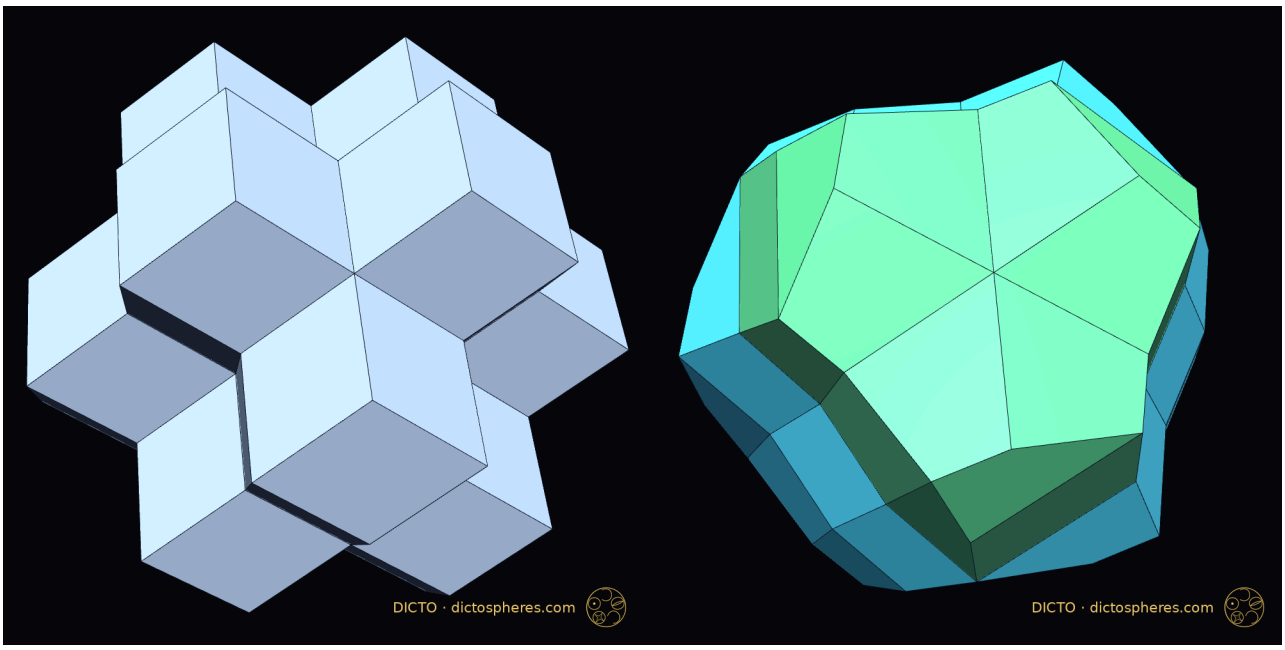


Figure 1. The plain thirteen rhombic dodecahedra, and the DICTO Spheroid-13 RD (thickness 0.15): the true rhombic dodecahedron at the centre gold, the two outer kinds green and blue.

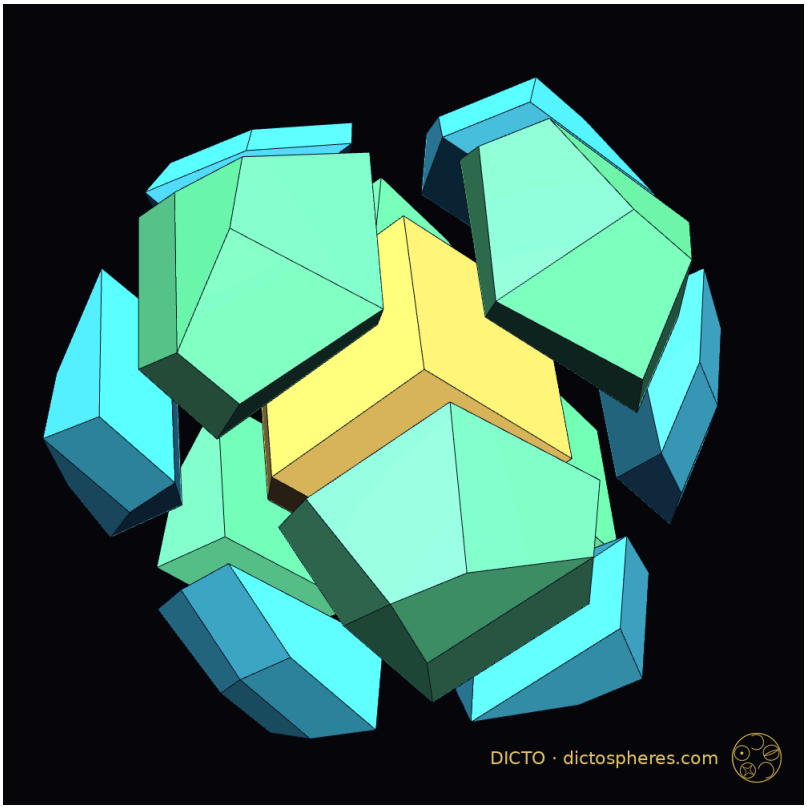


Figure 2. The block pulled apart, the rhombic dodecahedron at its heart.

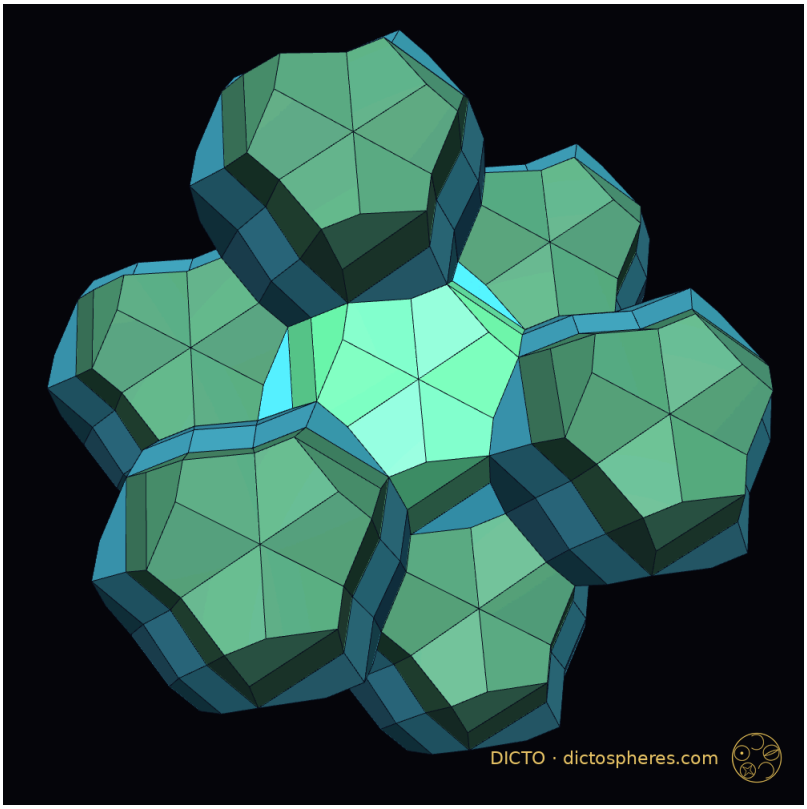


Figure 3. Seven blocks filling space, keyed (the six nearest neighbours dimmed).

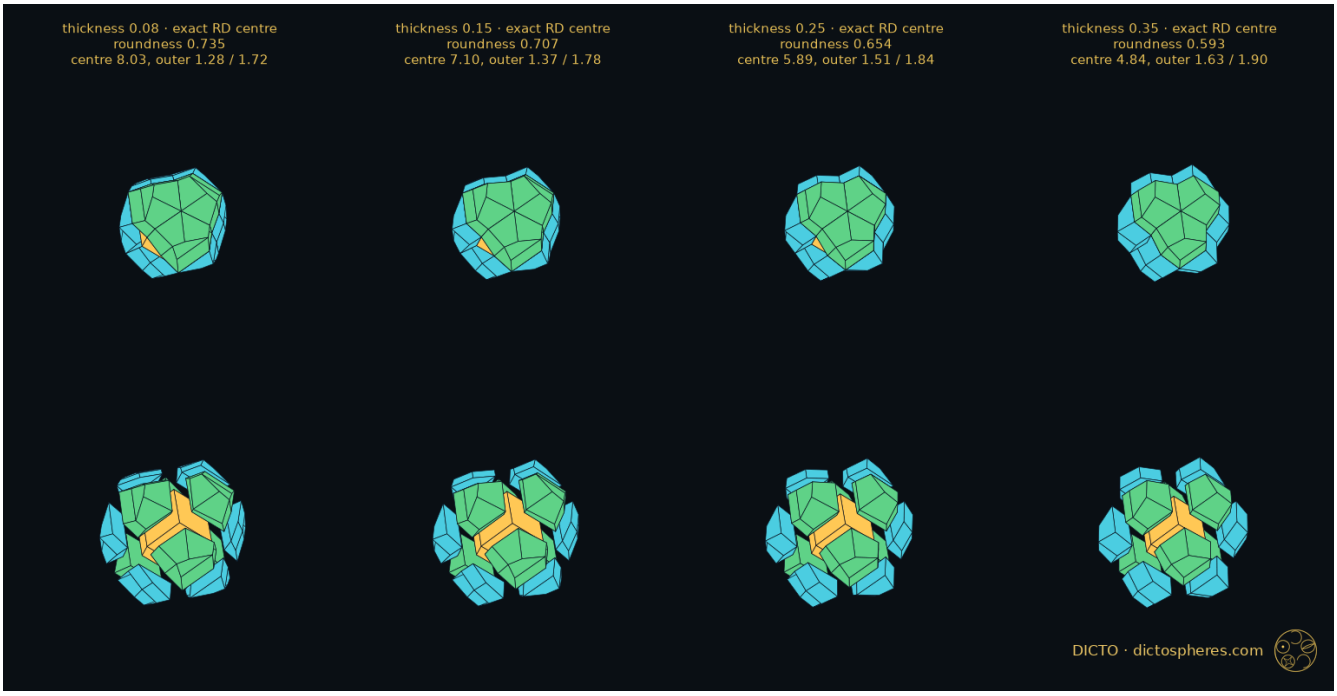


Figure 4. Minimum thickness 0.08, 0.15, 0.25 and 0.35, centre a true rhombic dodecahedron: whole (top) and pulled apart (bottom).

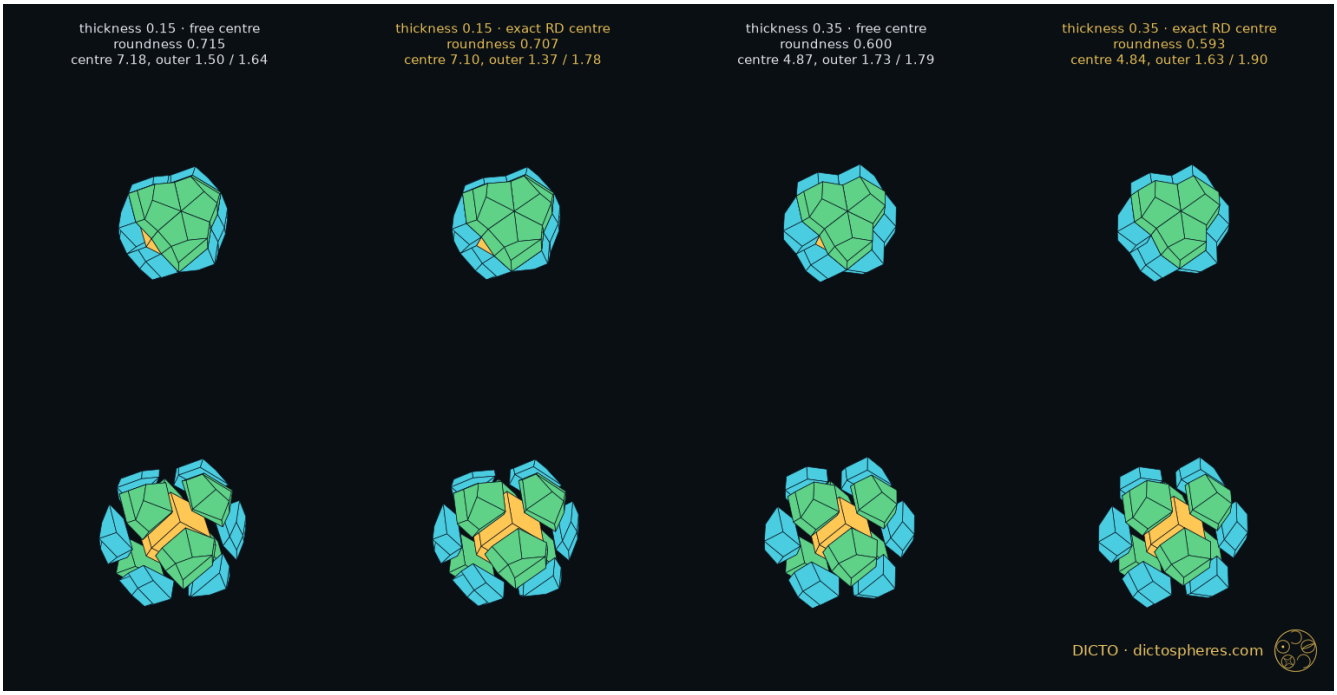


Figure 5. The free-centre variant beside the true rhombic dodecahedron centre, at thickness 0.15 and 0.35.

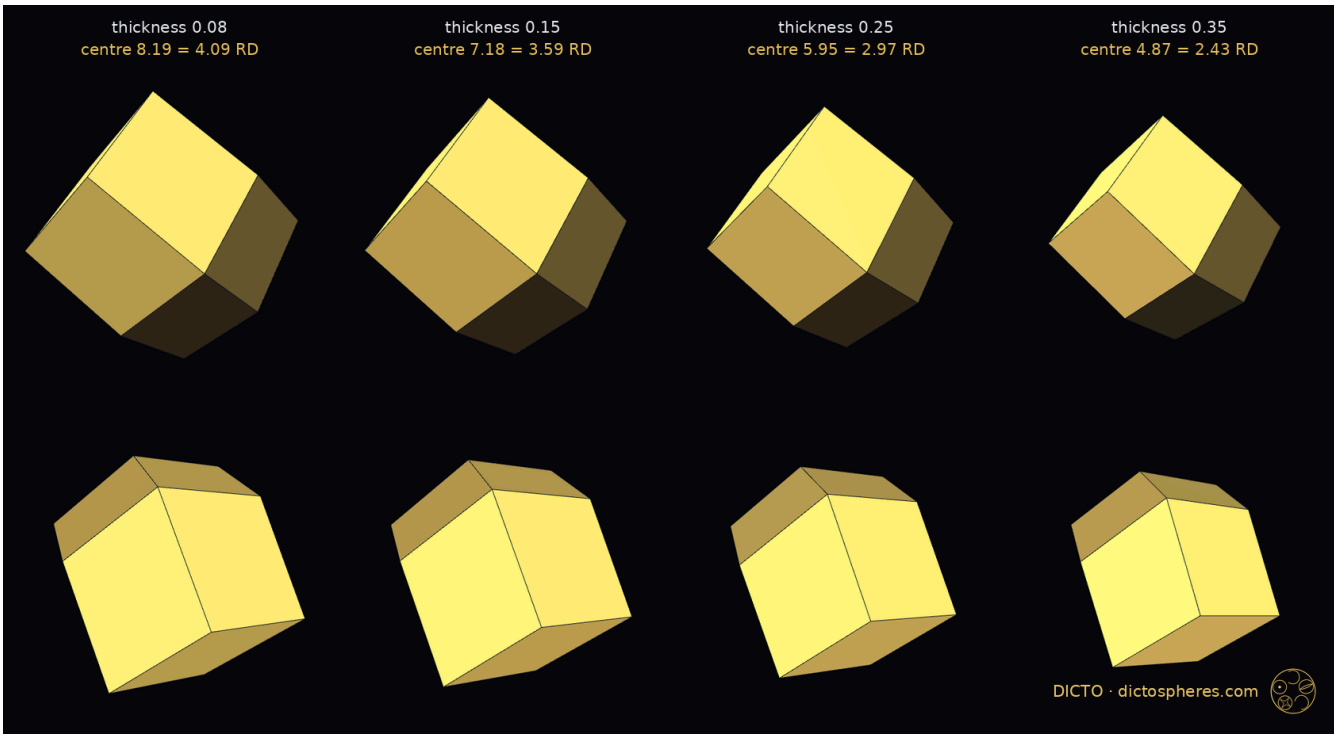


Figure 6. The variant's free centres at the four thicknesses, at true relative size: a graded set.

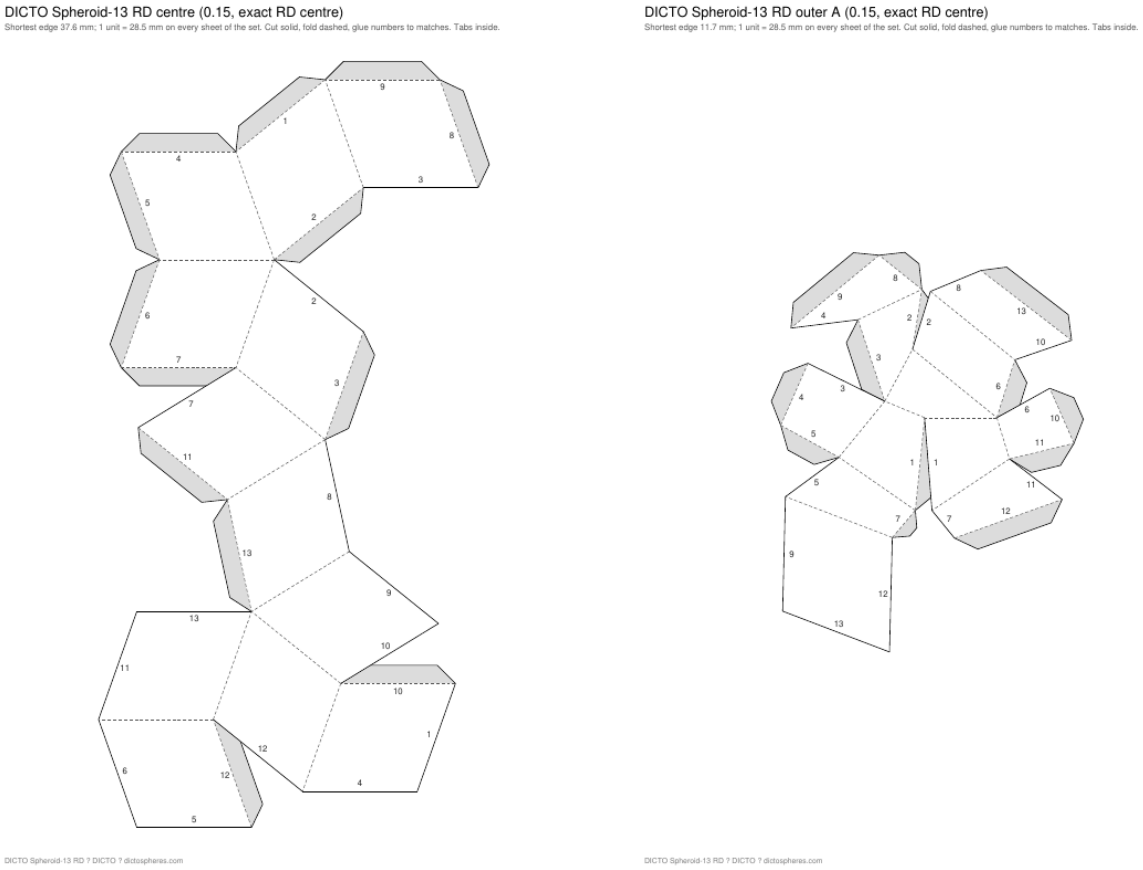


Figure 7. Two of the fifteen printable nets (thickness 0.15, one scale): the rhombic dodecahedron centre and outer piece A.

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